

ELEC5305 - Removing Unwanted Noise with Beamforming

Student Information

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Github Project Link: <https://github.com/Jdul9044/ELEC5305-Project-Jdul9044>

Project Overview

Currently in Digital Signal Processing (DSP), particularly in audio and acoustics the quality of a signal still remains to be an area of study and for future development. One way in understanding quality is to investigate its Signal to Noise Ratio (SNR). An audio recording with a high level of noise can contaminate the audibility and potentially make adjusting the gain redundant. In an amplifier if there is a low SNR in the input of the system as the gain is increased, the noise is also increased, hence making it hard to remove noise when it is already in the system. For higher grade audio equipment a lower noise floor is more attractive for studio application, however, for inexpensive products this may not be achievable. There are multitude techniques applied into optimising and improving SNR. One way is to “add some spatial information to the time frequency” which is achieved using multiple sensors, this is called beamforming [1]. For the basis of this project the intention is to demonstrate how a conventional beamformer can achieve SNR improvements in a noisy environment and possibly evolving the computation into an adaptive beamformer such as Minimum Variance Distortionless Response (MVDR).

Background and Motivation

The use of beamforming was a technique not well understood until the early 1900s by Guglielmo Marconi and Karl Ferdinand Braun and then later in the 1960s became introduced in sonar phased array systems [2]. After the 1960s adaptive beamforming methods were starting to be implemented which essentially are algorithms that adjust based on the environment through statistics and attempt to negate other interference or noise [3]. Although beamforming has mostly been a solved problem, standard conventional beamformers are still used today for their robustness. In addition, there also has been studies made on improving robustness as well on MVDR [3]. Due to the advancements of faster Analogue to Digital Converters (ADC) and easier to obtain electronics, access to digital beamforming has made both types of beamforming widely used today [4]. This DSP technique was chosen for exploration of complexity and for its wide use in audio application.

Proposed Methodology

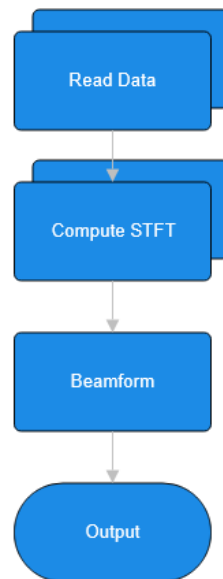
Tools and Platforms

For the main purpose of this project report the analysis will be conducted in Jupyter Labs with Python. Libraries such as Numpy, Scipy and Matplotlib will be used as the main packages.

However, generated simulated data will be produced using Python scripts. The main code will be developed for a post-processing environment.

DSP Technique

The overall analysis will consist of signal processing chain as followed:



Read Data - The raw data is initially read as an audio Pulse Code Modulation (PCM) file with a bit-depth of 16-bit is at least selected for minimal quantisation effects for each element of the array [5].

Compute Short-Time Fourier Transform (STFT) - For the analysis, beamforming in the frequency domain is selected as it is the fastest method for computation for post processing. A Discrete STFT is selected with a standard Hann window with an overlap of 50%. Window length and Discrete Fourier Transform (DFT) sizes will be selected for narrowband and wideband for performance metrics.

Beamform - A conventional beamformer will be implemented for the frequency domain that will be generalised to work with both linear and volumetric arrays. Further improvements may include MVDR.

Data Sources

1. Stage one (Simulation)
 - a. Simulate pure tonal at a particular azimuth and elevation for a linear array then a volumetric array.
2. Stage two (Online dataset)
 - a. Find raw array data for linear array.
 - b. Find raw array data for volumetric array.
3. Stage three (If time permits)
 - a. Test volumetric array with DIY hydrophone/microphones sensors.

Expected Outcomes

- Fully documented Jupyter Lab code with step by step processes and a script that generates simulated data.
- A performance comparison between standalone hydrophone/microphone and beamformed output results which will include SNR graphs and spectrogram display. May also include performance using a different number of elements.
- GitHub with documented and stored data sets.
- 8 channel DIY hydrophone/microphone array (if time permits).

Timeline

Week	Task
1-2	Start Initial Report and Literature Review
3-5	Begin Testing Simulated Data
6-9	Test Online Data and Record Findings
10-11	Potentially Testing Own Recorded Data and Optimisation
12-13	Report Completion

Reference

- [1] A. Hidri, S. Meddeb, A. Alaqeeli, and H. Amiri, "Beamforming techniques for multichannel audio signal separation," *J. Digit. Commun. Technol.*, 2012. [Online]. Available: <https://arxiv.org/pdf/1212.6080>
- [2], [3] A. M. Elbir, K. V. Mishra, S. A. Vorobyov, and R. W. Heath, "Twenty-five years of advances in beamforming: From convex and nonconvex optimization to learning techniques," *IEEE Signal Process. Mag.*, vol. 40, no. 4, 2023. [Online]. Available: [IEEE Xplore](#)
- [4] S. K. Sharma *et al.*, "Cognitive beamforming for spectral coexistence of hybrid satellite systems," in *Cooperative and Cognitive Satellite Systems*, 2015, ch. 13. [Online]. Available: ScienceDirect
- [5] R. Triggs, "Audio bit depth: Everything you need to know," *SoundGuys*, Dec. 17, 2024. [Online]. Available: [SoundGuys](#)